22 A 8.0 A current passes through a cylindrical copper wire with a diameter of 8.0 mm. The density of copper is 8960 kg m^{-3} and the mass of a single copper atom is 1 x 10^{-25} kg.

Assuming that there is one conduction electron for each copper atom, what is the drift velocity of the electrons in the wire?

A 2.8 x 10^{-6} m s^{-1} **B** 3.4 x 10^{-6} m s^{-1} **C** 1.1 x 10^{-5} m s^{-1} **D** 5.8 x 10^{-5} m s^{-1}

Ans:C

$I = nAvq$

$$v = \frac{I}{nAq} = \frac{8.0}{8960/(1\times10^{-25})\times\pi(4.0\times10^{-3})^2\times1.6\times10^{-19}} = 1.1\times10^{-5}\text{ m s}^{-1}$$

Distractors:
2.8 x 10^{-6}: Never convert diameter to radius.

23 Six resistors are connected in a circuit as shown below.